

What is Data Model?

Data models are made up of entities, which are the objects or concepts we want to track data about, and they become the tables in a database.

What is Database?

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. ... Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.

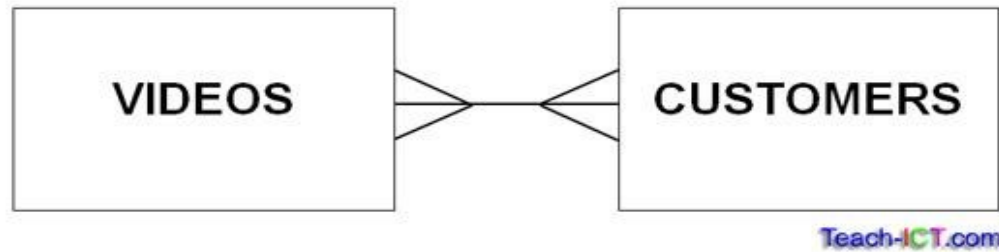
Why Database is Important?

- To manage large chunks of data .
- Accuracy .
- Security of data.
- Data Integrity .
- Organize data .

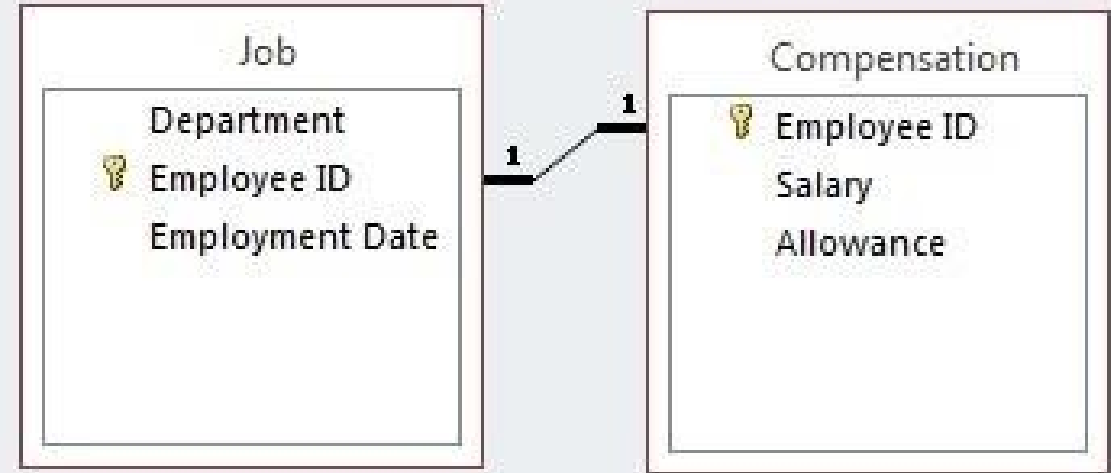
Types of Relationships

- Many to Many

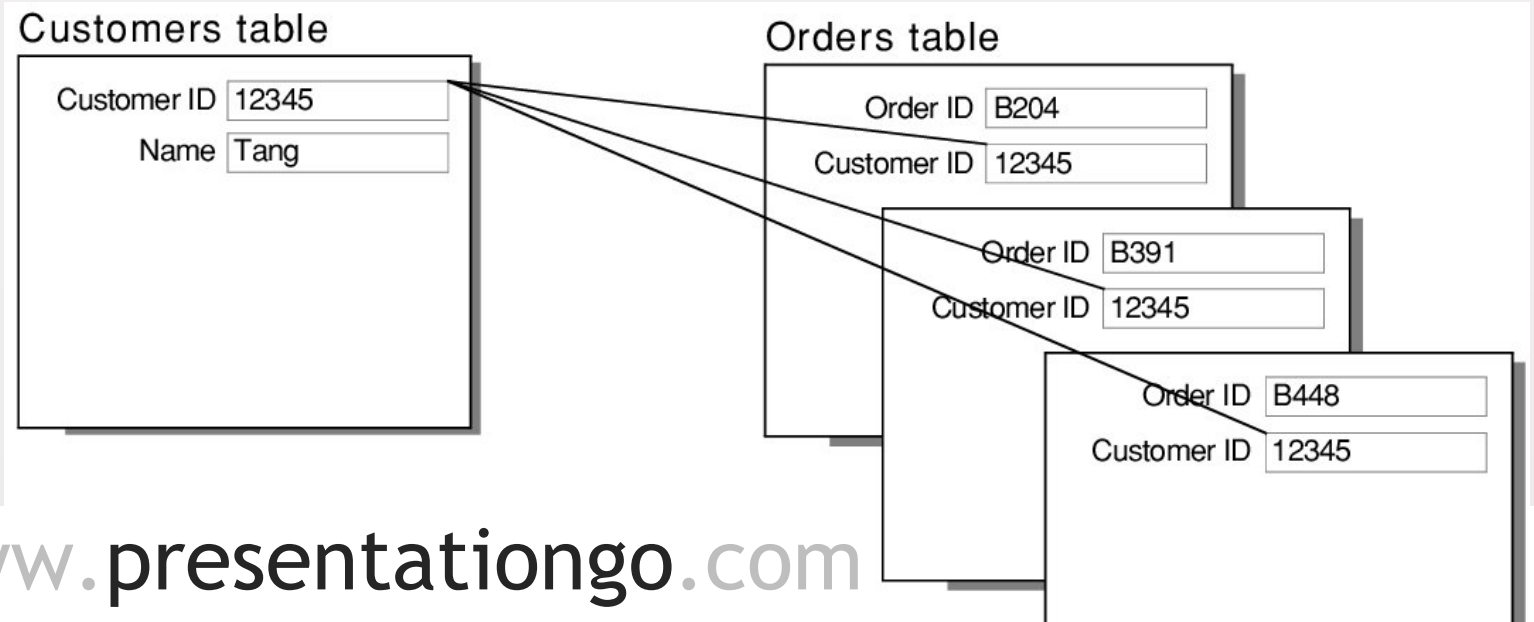
Many-to-many relationships



- One to One



- One to Many





What is Primary Key?

A primary key is a **column -- or a group of columns -- in a table that uniquely identifies the rows in that table.** For example, in the table below, Customer No, which displays the ID number assigned to different customers, is the primary key. CUSTOMERS. Customer No. First Name

The properties of each primary key column or columns are:

- It enforces uniqueness by not accepting any duplicate values.
- A primary key uniquely identifies each field.
- A table can only take one primary key.
- Primary columns have a maximum length of 900 bytes.
- A primary key column cannot accept null values.



What is dimension table?

A **dimension table** is a table in a star schema of a data warehouse. A **dimension table** stores attributes, or dimensions, that describe the objects in a fact ...



What is foreign Key?

A **foreign key** is a column or group of columns in a relational database table that provides a link between data in two tables.

What is fact table?

- A fact table or a fact entity is **a table or entity in a star or snowflake schema that stores measures that measure the business, such as sales, cost of goods, or profit.** ... Fact tables and entities use primary keys that are composite keys. A composite key is made up of a subset of other keys.

Register *

Description
_Date

tbl_data_RoadSegments *

Segment_Number
Segment_Validation_Date
Road_Number
Start_Chainage
End_Chainage
Asset_length
Start_Pt_X
Start_Pt_Y
End_Pt_X
End_Pt_Y
Start_Desc
End_Desc
Segment_Changed

Time
Dimension Table

Time_Key
Shift
Day
Day_of_Week
Month
Year

Test_Result
Fact Table

Time_Key
Test_Key
Patient_Key
Laboratory_Key
Test_Result
Remarks

Test
Dimension Table

Test_Key
Test_Name
Item_No
Analyzer_No
Test_Type
Reference_Value
Price

Patient
Dimension Table

Patient_Key
Patient_Name
Date_of_Birth
Gender
Address
Union/Ward
Thana
District

Laboratory
Dimension Table

Laboratory_Key
Laboratory_Name
Class
Lab_Address
Union/Ward
Thana
District



Examples of Data Model

